

N N Rajagopal

Final year student

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Profile Summary

Robotics and Automation undergraduate with hands-on experience in designing, developing, and deploying Autonomous Mobile Robots (AMRs) using ROS1 and ROS2. Experienced in building scalable navigation systems with Nav2, SLAM, AMCL, TF2, LiDAR, and waypoint-based autonomous navigation for real-world and simulated environments. Skilled in multi-robot systems, path planning, localization, Gazebo/RViz simulation, and robot commissioning, with practical expertise in integrating embedded hardware, sensors, and computer vision solutions for autonomous robotic applications.

Experience

Robot Operator Trainee (Intern)

07/2026 – Present

Spotless AI Robotics

Salem

- Deployed, operated, and maintained autonomous mobile robots (AMRs) for hospital disinfection using ROS2-based autonomous navigation.
- Configured and optimized ROS2 Nav2 navigation with Smac Planner and NavFn Planner, including SLAM, AMCL, LiDAR-based localization, and waypoint configuration.
- Troubleshoot and resolved mapping, localization, TF, navigation, and planner issues during real-world robot deployments.
- Collaborated with robotics engineers to validate autonomous and manual disinfection workflows, improve robot performance, and support field commissioning.
- Performed robot commissioning, system calibration, and functional testing before field deployment to ensure reliable autonomous operation.
- Collected and analyzed navigation performance data to optimize localization accuracy, path planning, and overall mission reliability in hospital environments.

CAD Designer Intern

02/2024 – 02/2025

Yoi Robotics LLP

Coimbatore

- Designed and modeled 10+ robotic components using SolidWorks and Fusion 360 for industrial automation systems.
- Reduced material usage by optimizing component geometry while maintaining structural integrity and fit tolerances.
- Contributed to assembly-level design reviews focusing on alignment, manufacturability, and real-world constraints.
- Collaborated with robotics and manufacturing teams through iterative design cycles from concept to production-ready parts.
- Prepared detailed engineering drawings, assembly documentation, and manufacturing-ready models following industry design standards.
- Performed design validation through interference checks, tolerance analysis, and iterative prototyping to improve assembly efficiency and product reliability.

Technical Skills

ROS / Navigation

ROS1 (Noetic), ROS2 (Humble), Nav2, AMCL, TF2, ros2_control, colcon, SLAM (gmapping, cartographer)

Programming

Python, C

CAD & Mechanical Design

SolidWorks, Fusion 360 (assembly design, component modelling)

Simulation

Gazebo Classic, Ignition Gazebo (Harmonic), RViz, Gazebo-ROS bridge

Embedded Systems & Hardware

Raspberry Pi, Arduino, Teensy Jetson, LiDAR integration, Odroid, STM32

DevOps / Tools

Linux (Ubuntu), Git/GitHub, SSH, VS Code, Platform IO

Projects

Swarm Robotics using ROS2

- Built and deployed a dual-robot system (bumpy5 & bumpy7) on Raspberry Pi using ROS2 Humble with Nav2.
- Resolved persistent TF2 frame issues with namespaced frames (bumpy7/base_footprint, bumpy7/odom) and fixed AMCL localization failures.
- Developed zone-based random patrol exploration script using NavigateToPose action with map-splitting logic for multi-robot coverage.
- Building a decentralized multi-robot coordination system using ROS2 with topic-based inter-robot communication.
- Implementing collaborative task allocation and formation control strategies for autonomous swarm behaviour.

Autonomous Mobile Robot – 250 kg Payload (ROS2)

- Developed a ROS2-based Autonomous Mobile Robot (AMR) capable of transporting payloads up to **250 kg** in industrial environments.
- Integrated the **Nav2** navigation stack with **AMCL**, **LiDAR**, and waypoint-based autonomous navigation for reliable indoor operation.
- Configured and optimized **Smac Planner**, **NavFn Planner**, and **DWB Controller** to improve path planning, obstacle avoidance, and navigation performance.
- Enhanced the ROS2 architecture with **namespaced multi-robot support**, scalable package organization, and launch configurations for deployment and testing.

Autonomous Mobile Robot – 20 kg Payload (ROS1)

- Designed and simulated a **20 kg payload** differential-drive Autonomous Mobile Robot using **ROS1 Noetic**.
- Implemented **2D SLAM (gmapping)**, localization, and autonomous navigation using the ROS Navigation Stack.
- Tuned global and local planner parameters to achieve smooth path following, accurate obstacle avoidance, and stable navigation.
- Validated robot performance in **Gazebo** and **RViz**, including mapping, localization, and waypoint-based navigation in simulated environments.

Cafe Butler Robot (ROS2)

- Developed a modular ROS2-based autonomous café service robot with separate packages for robot description, simulation, navigation, bring-up, and task execution.
- Designed and integrated the complete navigation stack using Nav2, LiDAR, AMCL, and waypoint-based autonomous table delivery.
- Built simulation and testing workflows in Gazebo with launch automation scripts for deployment, debugging, and workspace management.
- Implemented a scalable ROS2 workspace architecture, enabling efficient development, testing, and deployment of autonomous service robot functionalities.

Education

B.E. Robotics and Automation

2023 – 2027

Sri Ramakrishna Engineering College

CGPA: 8.05

Certifications

- Python Programming – Infosys Springboard
- SolidWorks – LinkedIn Learning
- MATLAB & Simulink Onramp – MathWorks